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Multi-Scale DCNN with Dynamic Weight and Part Cross-Entropy Loss for Skin Lesion Diagnosis

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Abstract: Accurately diagnosing skin lesion disease is a challenging task. Although present methods often use the multi-branch structure to get more clues, the rigescent methods of cropping zone and fusing branch results fail to handle the instability of the disease zone and the difference in branch results, which leads to improper cropping and degrades Deep Convolutional Neural Networks (DCNN)'s performance. To address these problems, we propose a Multi-scale DCNN with Dynamic weight and Part cross-entropy loss model (namely MDP-DCNN) to bootstrap skin lesion diagnosis. Inspired by the object detection method, the multi-scale structure adjusts the cropping position based on the Gradient-weighted Class Activation Mapping (Grad-CAM) center. It enables the model to adapt to the disease zone variety in position and size. The dynamic weight structure alleviates the negative influence of branch differences by comparing the grey-cropped zone and its CAM. Moreover, we also propose the part cross-entropy loss to deal with the over-fitting problem. This optimizes the non-targeted label to decrease the influence on other labels' stability when the prediction is wrong. We conduct our model on the ISIC-2017 and ISIC-2018 datasets. Experiments demonstrate that MDP-DCNN achieves excellent results in skin lesion classification without external data. Multi-scale DCNN with dynamic weight and part loss function verifies its advantages in enhancing diagnosis accuracy.

Key words: skin lesion; dynamic weight; dynamic cropping; part cross-entropy loss

1 Introduction

Skin cancer is one of the most common deadly diseases, mainly linked to exposure to ultraviolet radiation. According to data from the American Cancer Association (ACA), there will be 108 270 new cases of skin tumors in the USA in 2024[1]. Research indicates that early diagnosis of melanoma (MEL) can significantly improve the 5-year survival rate. Therefore, timely detection of MEL is essential to saving lives. Commonly, skin lesions are categorized into two main types: malignant and benign skin lesions. MEL is a type of malignant skin disease and typically manifests as an abnormal or irregular mole. Regular benign skin lesions include most melanocytic nevi (NV), commonly known as moles, Seborrheic Keratoses (SK), skin tags, cherry angiomas, and lipomas, among others. Unlike malignant lesions, benign skin lesions are usually symmetrical and well-circumscribed, have a uniform appearance, and are stable or grow slowly over time. Figure 1 provides an intuitive representation of skin lesion diversity and similarity. In medicine, experts diagnose skin diseases

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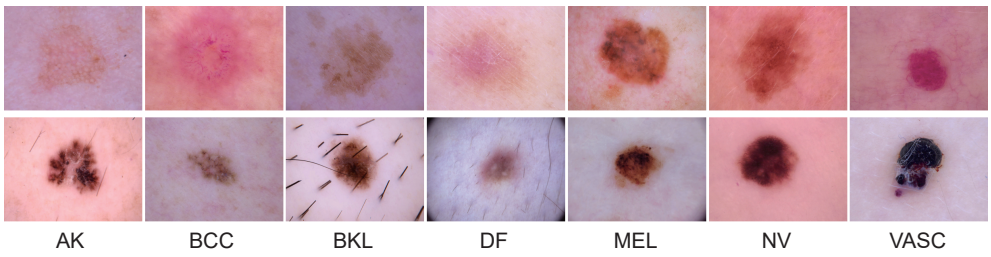


Fig. 1 Seven skin lesion classes in HAM10000 dataset, they are Actinic Keratosis (AK), Basal Cell Carcinoma (BCC), Benign Keratosis Lesions (BKL), dermatofibroma (DF), MEL, NV, and vascular lesions (VASC). The images in each row show the high variation within the same class, and the images in each column display the intra-class similarity.

through dermoscopy[2] and summarize a series of rule sets, such as the ABCD rule[3, 4] and 7-point checklist. However, these rules heavily rely on subjective evaluation and require skilled experience. Furthermore, the clinical features of SK have a non-uniform appearance, and MEL tends to occur near or inside SK. Some types of SK are similar to MEL, which poses an enormous challenge in diagnosing skin lesions[5].

Due to the excellent performance of computer science in various domains, Computer-Aided Diagnosis (CAD) has recently garnered researchers' attention for assisting dermatologists and improving diagnostic accuracy. To implement CAD in this area, certain challenges related to lesion disease characteristics must be addressed. For instance, (1) variations in the appearance of skin lesions caused by factors, such as hair, light, and blood vessels; (2) significant differences in the appearance of the same disease at different stages; and (3) the similarity between intra-class skin lesions and multi-class skin lesions.

Researchers have proposed various methods to address the aforementioned challenges, achieving remarkable improvements in the classification and segmentation of skin lesion diagnosis. Early studies focused on extracting handcrafted features as input for conventional classifiers[6, 7]. However, the lack of a unified image-capturing standard in many datasets limits the applicability of these approaches. Subsequently, automated skin lesion methods have been introduced[8–13]. Deep Convolutional Neural Networks (DCNNs) have demonstrated excellent performance in skin lesion classification or segmentation, eliminating the need for the complex feature extraction process in traditional methods and allowing more focus on exploring novel theories or techniques. In addition to developing new DCNN structures, introducing external datasets is a common strategy to enhance algorithm performance, as demonstrated in various task leaderboards, such as ImageNet[14]. Similarly, winners of competitions, like ISIC-2017, employ external datasets for improved performance[15–19]. However, accessing external datasets in the medical field is not always feasible. Furthermore, ensemble methods combining multiple models could further reduce recognition errors[5, 15]. In some sense, ResNet can be considered as an ensemble method[20], highlighting the effectiveness of such approaches. However, ensemble methods increase resource consumption and pose challenges in selecting proper weights or fusion methods. Inspired by fine-grained image categorization, such as fine-grained feature learning and part detection[21–23], researchers have transferred part detection into the skin lesion diagnosis domain to address inter-class similarity issues. In these methods, images are divided into several parts before applying part detection techniques and feeding them into Convolutional Neural Networks (CNNs). However, division without a target is ineffective and burdens CNN's recognition task. The introduction of Class Activation Mapping (CAM)[24], used as a visualization tool and attention mechanism[25], helps CNNs accurately locate key regions.

Additionally, to align with the model's characteristics, novel loss functions[26, 27] have been designed. Through an analysis of previous methods, it becomes evident that the fused method lacks flexibility in the ensemble approach. The weight of each branch should be individually dynamic based on its performance, and solely optimizing the loss of the targeted label is insufficient. To address these issues, we propose a Multi-scale DCNN with a Dynamic weight and Part cross-entropy loss model (namely MDP-DCM). We conduct extensive experiments on two public datasets, ISIC-2017 and ISIC-2018. The

results demonstrate that the novel model outperforms other works even without the use of an external dataset. The contributions of our work are summarized as follows:

(1) We introduce a multi-scale branch structure by adaptively locating the cropping center with Gradient-weighted CAM (Grad-CAM).

(2) We devise a dynamic branch weight mechanism by assessing its cropping effect.

(3) Part cross-entropy loss function is designed to overcome the possible over-fitting problem.

(4) We conduct extensive experiments on the ISIC-2017 and ISIC-2018 datasets.

The remainder of the paper is organized as follows: Section 2 provides an overview of related works in skin lesion classification. In Section 3, we present the detailed pipeline of the MDP-DCNN. The experiments and results are presented in Section 4. An ablation study is conducted in Section 5. Section 6 is dedicated to discussing and analyzing the results. Finally, Section 7 summarizes the contents of the paper and outlines plans for future research.

2 Related Work

A substantial number of works have studied skin lesion classification in recent years. Initially, some methods naturally attempted to mimic the human mechanism for discriminating diseases. They extracted features based on skin lesion attributes such as texture, color, and asymmetry[28, 29], subsequently feeding these features into a classifier[30, 31] to obtain the final result. However, this feature extraction process is time-consuming, and the diversity of diseases makes predictions unreliable. Since the advent of DCNN, it has achieved remarkable success in computer vision and garnered much attention. The end-to-end training pattern has significantly simplified cumbersome handcrafted feature extraction processes. Numerous works have enhanced DCNN performance through various approaches, including extra datasets, ensemble methods, key zone location, feature fusion, attention mechanisms, data augmentation, and novel loss functions. The use of external datasets enhances models, placing them at the top of leaderboards and even achieving state-of-the-art results[32]. For instance, in the ISIC-2017 and ISIC-2018 Challenges, all the top-5 works incorporate extra datasets. Ge et al.[33] proposed a DCNN model with a saliency feature descriptor capable of capturing discriminative features for skin lesion classification across two modalities. Zhang et al.[13] introduced an ARL-CNN model that combines residual learning and a novel attention-learning mechanism assisted by CAM, demonstrating better disease zone localization and higher accuracy without any extra dataset. Gessert et al.[34] employed a patch-based attention architecture to utilize high-resolution information and applied multiple techniques to address issues like class imbalance. Tang et al.[5] utilized an ensemble of three branches to gather global and local image information through color constancy transformation, asserting that this new ensemble learning strategy enhances the ability to distinguish skin cancers. Xie et al.[10] jointly combined segmentation and classification to boost performance in both tasks. Liu et al.[8] leveraged fine-grained image classification, using multi-scale branches and margin loss for more accurate discrimination. In addition to discovering more clues, some researchers realize the importance of data augmentation. They implement Generative Adversarial Networks (GAN)[35] to generate samples and address the class imbalance problem. Bissoto et al.[36] summarized GAN-based data augmentation on medical images and conducted comparative experiments using SPADE[37], pix2pixHD[38], PGAN[39], and StyleGAN2[40]. They found that the performance of different GANs fluctuates across datasets, but translation-based GANs tend to work better than noise-based GANs. Due to the challenging training process of GAN-based models, some researchers have turned to the diffusion model to improve skin lesion diagnosis[41]. Designing proper loss functions to adapt to different models is one of the most commonly used techniques, facilitating training based on emphasized metrics. Song et al.[11] employed a three-phase joint training strategy to ensure efficient feature learning with a loss function based on focal loss and the Jaccard distance. Yao et al.[9] proposed a multi-weighted new loss method, claiming that it can simultaneously address data imbalance and outlier problems.

3 Method

The proposed method aims to dynamically locate the essential zone and utilize multiple branches to acquire more discriminative information and alleviate intra-similarity problem. For the high variation, multiple

branch structures with more parameters can include enough potential latent space and the part cross-entropy loss also makes probability distribution more widespread to alleviate the high variation within the same class. To some extent, the multiple-branch approach is similar to the patch-based method that divides the input image into many patches. However, such division could compromise spatial information. The MDP-DCNN pipeline is presented in Fig. 2. Similar to the skin lesion diagnosis process referring to the 7-point checklist and ABCD rules, the MDP-DCNN framework comprises three stages: extracting discriminative information, synthetically using multiple-branch advantages for prediction, and optimization. In the first stage, MDP-DCNN employs three branches with different-resolution images to obtain features. The low-resolution branch is to extract global information, such as the outline, shape, or symmetry. Grad-CAM, inserted in the last layer, collects each pixel's gradient linked to the target label and produces the CAM. After locating the center on the CAM, two high-resolution images are cropped from the original image and serve as the inputs for the other branches. The high-resolution branches focus on more detailed information, such as veins and tiny structures. However, high resolution sometimes brings certain negative information due to the loss of much global information, so it is necessary to assess the performance of the high-resolution branches. Thus, in the second stage, we design the weight block to dynamically manipulate the branch's weight. The inputs of the weight block are its branch's CAM and the corresponding reversed-grey image. The weight block consists of two dilated convolutions with Global Average Pooling (GAP) and a sigmoid function producing the weight factor. When the branch performs poorly, there will be an obvious discrepancy between the CAM and the reversed-gray image; the weight block tries to learn this pattern to change the branch weight. In the third stage, all three branch results are fused. Finally, the model gives its prediction, and errors will backpropagate with the part cross-entropy loss function. Significantly, the CAM at the initial training stages is poorly obtained, so the high-resolution branches do not work in the first several stages. Considering the performance of existing structures, MDP-DCNN adopts three pre-trained sub-models, ResNet[42], MobileNetV2[43], and FcaNet[44], as the backbones. These three models form four combination structures. The first three structures are composed of a single model. The last structure, also named the fused model, consists of the previous three models. ResNet searches for whole feature extraction, MobileNetV2 aims to obtain middle-level features, and

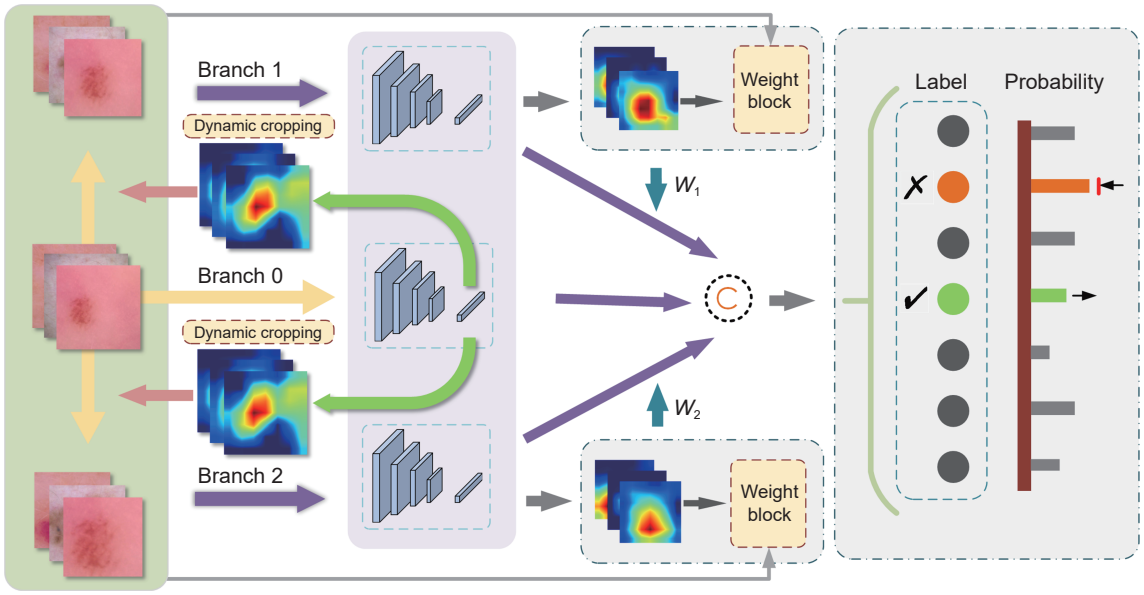


Fig. 2 Illustration of the proposed MDP-DCNN model. The proposed model comprises three branches and utilizes Grad-CAM to capture information at different scale levels. The CAM of the main branch locates the vital zone and crops two higher-resolution images to serve as the input for other branches. The weight block checks the effect by analyzing its CAM and reversed-grey image, producing branch weight (W_1 and W_2). Finally, the results of the three branches are fused (C), and the prediction is output using the part cross-entropy loss function.

FcaNet with frequency channel discriminates tiny variations. Finally, the end of these three branches forms a single model.

3.1 Cropping scheme

CAM[24] serves as the visualization tool and is currently used to assist the model in locating the key zone. The original CAM requires cumbersome modification and necessitates altering the pre-trained model structure, which can impair its performance. Thus, this work adopts Grad-CAM[45] as the zone selector without any modification. The cropping scheme is shown in Fig. 3, where Grad-CAM extracts the last layer's gradient information with the targeted label as the CAM. Then, the convolutional operation is applied to the CAM to obtain the central point (C_x, C_y). Given the width and height of the expected cropping part as (w, h), the coordinates of the left upper corner (C_{x1}, C_{y1}) and the bottom right corner (C_{x2}, C_{y2}) are calculated as follows:

$$\begin{aligned}&(C_{x1}, C_{y1}) = (\text{Max}\,(0, C_x - h/2), \text{Max}\,(0, C_y - w/2)),\\ &(C_{x2}, C_{y2}) = (\text{Min}\,(I_w, C_x + h/2), \text{Min}\,(I_h, C_y + w/2)),\\ &\quad \text{s.t.},\ C_{x2} \leftarrow I_h, \text{ if } C_{x1} = 0;\\ &\qquad C_{x2} \leftarrow I_h - h, \text{ if } C_{x2} = I_h;\\ &\qquad C_{y2} \leftarrow I_w, \text{ if } C_{y1} = 0;\\ &\qquad C_{y2} \leftarrow I_w - w, \text{ if } C_{y2} = I_w\end{aligned} \tag{1}$$

where I_w and I_h are the width and height of the original image.

3.2 Weight block

Certainly, the branch results are different. Weight, vote, and stack methods are the most commonly used combination mechanisms for obtaining predictions. However, the weight and stack methods overlook branch dynamic performance and assign a fixed weight to each branch. The weight value, calculated through limited experiments or set the same for all branches, is more prone to overfitting. Some works applied the margin loss[8, 23] to adjust the importance of multiple branches. However, the difference in branch values in the margin loss is limited by the margin, potentially degrading the multiple branches model into a single branch. To address these drawbacks, we propose a weight block by fusing the CAM and the reversed-gray image to dynamically adjust the weight. The details are listed in Fig. 4. Initially, reversely graying the cropping image, then, adding the CAM and the gray image together. The reversed-gray operation is to make it comparable between CAM and input image. Meanwhile, it also reduces the workload of the weight block. We utilize dilated convolution to extract outline information for enhancing the receptive range. Finally, through the synthetic computation of the input, a sigmoid function produces the weight.

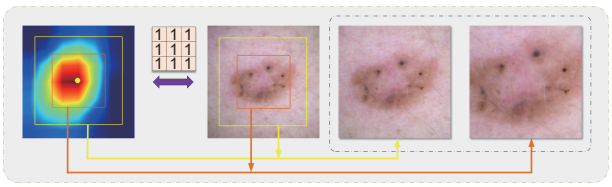


Fig. 3 Cropping scheme. Applying a convolutional operation on the CAM and locating the most supportive pixel as the center, then cropping the relative zone based on the center. The orange and yellow rectangle sizes are 336 pixel × 336 pixel, and 224 pixel × 224 pixel, respectively.

3.3 Part loss function

In the classification task, for the sake of effectiveness and simplicity, cross-entropy loss is the most commonly used loss function, which solely reduces the target loss. According to the theory of softmax and the prediction rule, the final result depends on the prediction probability for each class. Li et al.[46] proposed dual cross-entropy for fine-grained classification. Their method format is similar to the combination of full cross-entropy loss (full loss) and cross-entropy loss.

Assume $D = \{(x_1, y_1), \ldots, (x_i, y_i), \ldots, (x_N, y_N)\}$ is the training dataset with N instances, where y_i, a one-hot vector, is the label of x_i. $Z_{x_i}^j$ is the predicted value in the j-th class. Suppose p_i is the probability vector

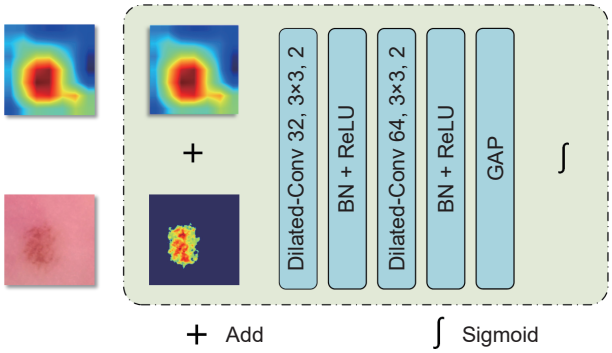


Fig. 4 Illustration of weight block. In the weight block, we gray the cropped image reversely, combine the CAM and the cropped image as the inputs of two dilated convolutions, and obtain the weight factor by the sigmoid function. The parameters of dilated convolutions represent out-channel number, kernel size, and dilation rate. BN denotes batch normalization.

containing C elements, element (p_{ij}), the probability that sample x_i is assigned to the j-th class, $j \in \{1, 2, \ldots, C\}$, is calculated according to

$$p_{ij} = \frac{e^{Z_{x_i}^j}}{\sum_{i=1}^{C} e^{Z_{x_i}}} \tag{2}$$

The prediction label of x_i is p_i^L. When the prediction is right, p_i^L is the targeted label, otherwise, p_i^L is the label of the largest probability class. $\mathcal{L}_{\text{CE}}$ means cross entropy loss,

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N}\sum_{i=1}^{N}(y_i^{\text{T}} \log(p_i)) \tag{3}$$

In Eq. (3), only the target error can backpropagate. Whether or not the prediction is correct, the values of p_i and $Z_{x_i}^{y_i}$ continue to grow. The non-targeted errors should also be noticed.

Full cross-entropy loss cares for all class probabilities; part cross-entropy loss (part loss) will additionally optimize the largest probability label when the prediction is wrong,

$$\mathcal{L}_{\text{full}} = -\frac{1}{N}\sum_{i=1}^{N}(y_i^{\text{T}} \log(p_i) + (1 - y_i^{\text{T}}) \log(1 + p_i)) \tag{4}$$

$$\mathcal{L}_{\text{part}} = -\frac{1}{N}\sum_{i=1}^{N}(y_i^{\text{T}} \log(p_i) + \beta \cdot (1 - y_{ij}) \log(1 + p_{ij})) \tag{5}$$

$$\beta = \begin{cases} 0, & p_i^L = y_i; \\ \lambda, & p_i^L \neq y_i \end{cases} \tag{6}$$

where y_{ij} is the j-th element in y_i, and j is the index of the largest probability class.

In Fig. 5, there are seven classes, and the 4-th class is the real label. Cross-entropy loss aims to enhance the probability (P) of 4-th class untilite is the largest, full loss intends to increase the target label probability and decrease other label probabilities. Part loss is different from the previous loss mechanisms. It increases the target label's probability and decreases the largest probability when the prediction is wrong.

4 Experiment and Result

In this study, we select two of the most used datasets in skin lesion diagnosis, ISIC-2017 and ISIC-2018.

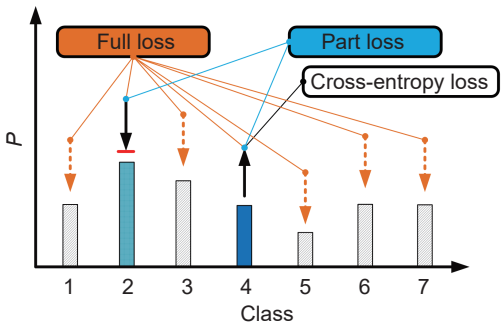


Fig. 5 Three types of loss function. There are seven classes and the arrow on each class represents the optimization direction. There are three kinds of loss functions, cross-entropy loss, part loss, and full loss.

4.1 ISIC-2017 dataset

The ISIC-2017 dataset originates from the 2017 International Skin Imaging Collaboration (ISIC) Challenge Task 3. It comprises 2000 training images, 150 validation images, and 600 test images. Table 1 presents its class distribution. Additionally, ISIC-2017 exhibits various image sizes, as illustrated in Fig. 6. To eliminate disturbances from boundaries and standardize the size, we perform center cropping covering 80% of the image area and resize the image to 600 pixel × 450 pixel. It is noteworthy that the ratios of width and height in the training set and test set are nearly the same. The cropping operation does not affect the image quality.

4.2 ISIC-2018 dataset

In the ISIC-2018 Challenge, the dataset adopted is HAM10000[47]. HAM10000 includes dermatoscopic images from different populations acquired and stored by various modalities. It consists of 100 015 dermatoscopic images released as a training set for academic machine learning purposes and is publicly available through the ISIC archive. There are seven categories, shown in Table 1.

4.3 Implementation details

The MDP-DCNN is trained for 200 epochs with an input size of 224 pixel × 224 pixel. For the ISIC-2018 dataset, the images are cropped from the original

Table 1 Class distribution of ISIC-2017 and HAM. There are seven types in HAM and three types in ISIC-2017.

Dataset	MEL	NV	BCC	AK	BKL	DF	VASC	SK
ISIC-2017	520	1843	0	0	0	0	0	386
HAM	1113	6705	514	327	1099	115	142	0

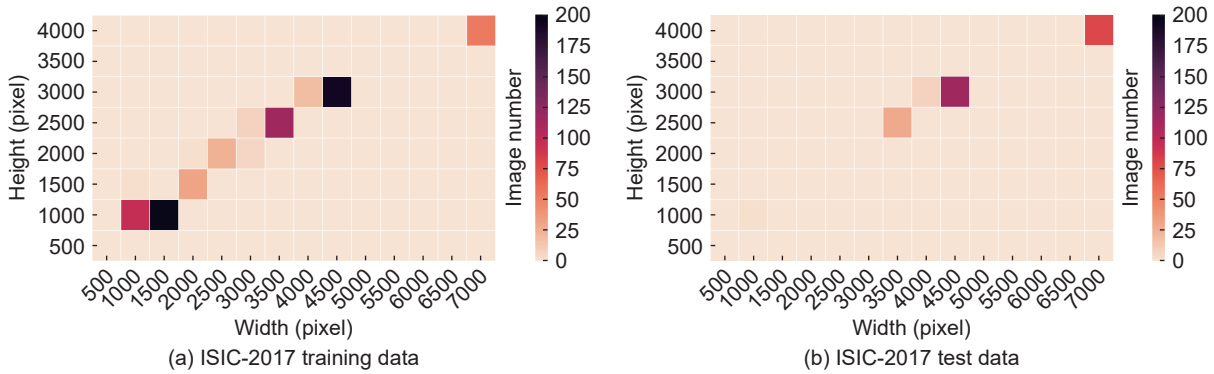


Fig. 6 Size distribution of ISIC-2017 dataset for training and test. Nearly all the images are in the same ratio of width and height.

images by 448 pixel × 448 pixel and then resized to 224 pixel × 224 pixel. The dataset is divided into a training set (80%) and a test set (20%), and the model is trained using five-fold cross-validation. For the ISIC-2017 dataset, cropping a certain zone with the original resolution is challenging due to various image sizes. Assuming one image size is $W \times H$ ($W > H$), the cropping size is $H \times H$, and then it resized to 448 pixel × 448 pixel. Moreover, online data augmentation is applied, including random rotation from 0 to 10 degrees, shear from 0 to 0.1 radians, and horizontal and vertical flips. Due to the poor ability of the Grad-CAM at initial steps, the high-resolution branches start to work until the 10-th epoch. SGD[48] with a batch size of 64 is employed to optimize the model. The initial training learning rate is set to 0.01, with a decay rate of 0.05, using the part cross-entropy loss function with $\lambda = 1$. The experiment is conducted on a platform with two NVIDIA Tesla V100 GPUs and 64 GB of memory.

4.4 Evaluation metrics

In this work, accuracy (Acc), balanced accuracy (BAcc), sensitivity (Sen), specificity (Spe), and Area Under the receiver operating characteristic Curve (AUC) are adopted to evaluate the model. Note that the ISIC-2017 and ISIC-2018 challenges use the average AUC to rank the classification effect. The calculation methods are as follows:

$$\mathrm{Acc} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{FP}+\mathrm{TN}+\mathrm{FN}} \times 100\% \tag{7}$$

$$\mathrm{BAcc} = \frac{1}{M}\sum_{i=1}^{M} \mathrm{Acc}_i \times 100\% \tag{8}$$

$$\mathrm{Sen} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \times 100\% \tag{9}$$

$$\mathrm{Spe} = \frac{\mathrm{TN}}{\mathrm{TP}+\mathrm{FN}} \times 100\% \tag{10}$$

$$\mathrm{AUC} = \int_0^1 t_{\mathrm{PR}}\,(f_{\mathrm{PR}})\mathrm{d}\,(f_{\mathrm{PR}}) \times 100\% \tag{11}$$

where TP, TN, FP, and FN indicate true positive, true negative, false positive, and false negative, respectively, t_{PR} is the true positive rate, f_{PR} is the false positive rate, and M is the number of classes.

4.5 Comparing with state-of-the-art methods on the ISIC-2017 dataset

We compare our proposed model to recent works, including those from the ISIC-2017 Challenge. Table 2 presents a comparison with competitive models, such as the Attention Residual Learning Convolutional Neural Network (ARL-CNN)[13], a Synergic Deep Learning (SDL) model[49], Mutual Bootstrapping Deep Convolutional Neural Networks (MB-DCNN)[10], Global-Part Convolutional Neural Network (GPCNN)[5], Semi-Supervised Adversarial Classification model (SSAC)[50], and Image Resolution for Transfer Learning-based model (IRTL)[51]. Additionally, we include five top-ranking methods from the ISIC-2017 skin lesion classification leaderboard[15–19]. Furthermore, we adopt three pre-trained models, forming four combinations—Ensemble, MobileNetV2, ResNet50, and FcaNet—to classify Melanoma and Keratosis. The comparison is evaluated using commonly used metrics. In Table 2, our MobileNetV2 achieves the highest average AUC (93.1%). For Keratosis classification, it demonstrates values of 76.2%, 93.3%, 90.7%, and 96.0% for Sen, Spe, Acc, and AUC, respectively. In Melanoma classification, MobileNetV2 shows values of 57.3%,

Table 2 Skin lesion classification performance comparison on the ISIC-2017 testing set.

Method	External data	Keratosis classification				Melanoma classification				Average AUC (%)
		Sen (%)	Spe (%)	Acc (%)	AUC (%)	Sen (%)	Spe (%)	Acc (%)	AUC (%)	
ARL-CNN[13]	1320	87.8	86.7	86.8	95.8	65.8	89.6	85.0	87.5	91.7
SLD4 [49]	1320	–	–	92.5	95.8	–	–	88.8	86.8	91.3
Method in Ref. [15]	1444	97.8	77.3	80.3	95.3	73.5	85.1	82.8	86.8	91.1
Method in Ref. [16]	900	17.8	99.8	87.5	96.5	10.3	99.8	82.3	85.6	91.0
Method in Ref. [17]	7544	35.6	99.0	89.5	94.3	54.7	95.0	87.2	87.4	90.8
Method in Ref. [18]	1600	58.9	97.6	91.8	92.1	42.7	96.3	85.8	87.0	89.6
Method in Ref. [19]	1341	55.6	97.6	91.3	93.5	35.0	96.5	84.5	83.5	88.6
MB-DCNN[10]	1320	84.4	94.3	93.0	97.3	72.7	91.5	87.8	90.3	**93.8**
GP-CNN-DTEL[5]	0	72.2	97.3	93.5	96.0	37.6	96.5	85.0	89.1	92.6
SSAC[50]	1320	88.9	91.6	91.2	95.9	55.6	90.3	83.5	87.3	91.6
IRTL[51]	0	–	–	–	–	–	–	–	–	92.9
Ours (Ensemble)	0	71.1	92.4	89.2	94.0	57.2	92.5	85.6	90.8	92.4
Ours (MobileNetV2)	0	76.2	93.3	90.7	96.0	57.3	91.1	84.5	90.2	93.1
Ours (ResNet50)	0	70.3	91.6	88.4	93.5	53.0	91.8	85.0	89.7	91.7
Ours (FcaNet)	0	68.9	91.0	87.7	93.0	56.4	90.6	83.9	89.0	91.0

Note: Number in bold means the best result, the red number indicates the second best result. Follow-up tables are the same.

91.1%, 84.5%, and 90.2% for Sen, Spe, Acc, and AUC, respectively. Compared with previous works, MobileNetV2 achieves the second-highest AUC. Its average AUC is 0.5%, and 1.4% higher than GP-CNN-DTEL and ARL-CNN, and 0.7% lower than MB-DCNN. External data and segmentation tasks could enhance the performance of MB-DCNN. Moreover, in previous studies, Melanoma disease, being similar to Nevus, has the largest number, which could lead to better performance in Keratosis classification compared to Melanoma classification. Additionally, our models exhibits more balanced results in terms of sensitivity in Melanoma classification compared to GP-CNN-DTEL.

4.6 Comparing with state-of-the-art methods on ISIC-2018 dataset

The classification task on the ISIC-2018 dataset poses increased complexity compared to the ISIC-2017 dataset due to a larger number of classes, resulting in variations in evaluation metrics. Key metrics for evaluation include BAcc and Sensitivity Melanoma (Sen. MEL), aligning with prior studies. In Table 3, our models are compared with three top-ranking results from the legacy leaderboard, five top-ranking results from the current leaderboard, and five state-of-the-art works. Notably, many models in the leaderboard leverage external datasets to enhance performance. The Ensemble model achieves a BAcc of 89.1%, Sen. MEL of 82.0%, securing the second and third positions compared to models in the leaderboard. Given that Melanoma is the most critical disease among the seven classifications, a high Sen. MEL value indicates the model's effectiveness in minimizing misdiagnosis. Additionally, other models in our study perform well, with MobileNetV2 outperforming ResNet and FcaNet. Interestingly, the frequency mechanism in FcaNet does not exhibit advantages over other approaches.

4.7 Loss function

We analyze the effects of different loss functions, as illustrated in Fig. 5, and summarized the results in Tables 4 and 5. Our findings indicate that the part cross-entropy loss function consistently outperforms the other two alternatives across the four models, although all three loss functions yield competitive results. The cross-entropy loss solely considers the probability of the target label, neglecting potential inter-label influences, which may result in overfitting. While optimizing for all labels might seem like a suitable choice to ensure alignment with the target label, the results show that full cross-entropy produces lower performance than the other methods. We attribute this to the more stringent constraint imposed by full cross-entropy, indicating a propensity for overfitting. In contrast, part cross-entropy operates specifically on the prediction label when the predicted result is incorrect, striking a balance between

Table 3 Performance of different methods. Results in Rows 1–3 and 4–8 are from the ISIC-2018 Challenge winners' legacy leaderboard and the current ISIC-2018 Challenge live leaderboard, respectively. The rest results are the recent works and our proposed methods.

(%)

Method	External data	Sen. MEL	Spe	AUC	BAcc
Method in Ref. [9]	Yes	76.0	83.3	98.3	88.5
Method in Ref. [34]	Yes	80.1	98.4	98.7	85.6
Method in Ref. [9]	No	70.2	98.0	97.8	84.5
Method in Ref. [9]	Yes	77.8	98.1	98.2	**89.5**
Method in Ref. [9]	No	81.3	97.5	98.3	88.6
Method in Ref. [52]	Yes	58.5	99.2	97.9	87.4
Method in Ref. [9]	No	86.0	98.0	95.3	87.3
Method in Ref. [9]	Yes	83.0	97.6	97.6	86.6
CL[53]	No	–	–	–	86.5
MWNL-CLS[9]	No	81.9	98.0	98.5	87.5
MECNN[8]	No	76.0	98.1	–	87.0
PROCO[54]	No	–	–	–	88.7
CI-Net[55]	No	–	98.0	98.5	87.3
Ours (Ensemble)	No	82.0	98.2	98.0	89.1
Ours (MobileNetV2)	No	77.0	97.6	98.0	88.4
Ours (ResNet50)	No	71.0	97.5	97.6	87.9
Ours (FcaNet)	No	68.0	97.5	97.4	86.8

Table 4 Loss function influence analysis on ISIC-2017.

(%)

Method	Loss			Keratosis classification				Melanoma classification				Average AUC
	Part	Full	Cross	Sen	Spe	Acc	AUC	Sen	Spe	Acc	AUC	
Ensemble	✓	×	×	71.1	92.4	89.2	**94.0**	57.2	92.5	85.6	90.8	**92.4**
Ensemble	×	✓	×	68.9	91.4	88.0	92.8	43.6	93.2	83.5	**91.0**	91.9
Ensemble	×	×	✓	70.3	92.1	88.8	93.3	51.3	91.7	83.8	89.0	91.5
MobileNetV2	✓	×	×	76.2	93.3	90.7	**96.0**	57.3	91.1	84.5	**90.2**	**93.1**
MobileNetV2	×	✓	×	60.0	90.6	86.0	91.3	55.6	89.4	82.8	89.3	90.3
MobileNetV2	×	×	✓	77.8	91.4	89.3	92.9	59.8	89.9	84.0	89.7	91.3
ResNet50	✓	×	×	70.3	91.6	88.4	**93.5**	53.0	91.8	84.3	89.7	**91.7**
ResNet50	×	✓	×	54.4	95.1	89.0	93.0	46.2	91.9	83.0	90.0	91.5
ResNet50	×	×	✓	57.8	92.9	87.7	92.6	53.0	91.5	84.0	**90.5**	91.6
FcaNet	✓	×	×	68.9	91.0	87.7	93.0	56.4	90.6	83.9	89.0	**91.0**
FcaNet	×	✓	×	57.8	91.4	86.3	91.0	51.3	90.7	83.0	**89.4**	90.5
FcaNet	×	×	✓	69.0	91.0	87.7	**93.3**	52.1	89.4	82.2	88.5	90.9

traditional cross-entropy and full cross-entropy. This characteristic contributes to its superior performance in our experiments.

4.8 Fused method

In addition to utilizing external datasets, fused techniques are commonly employed to enhance the model performance. In our study, we investigate four fusion methods: the result fusion method (weight method, stack method, vote method) and feature fusion method. The weight method incorporates a weight block to train suitable weights based on CAM and gray image inputs. Subsequently, it aggregates all results with the weights. The stack method directly combines the results from all three branches, treating each branch equally. The vote method determines predictions based on a majority mechanism. The feature fusion method concatenates all branch features together. We apply these four fusion methods to the ensemble model with three loss functions, as shown in Tables 6 and 7. The

Table 5 Loss function analysis on ISIC-2018.

(%)

Method	Part loss	Full loss	Cross loss	Sen. MEL	Spe	AUC	BAcc
Ensemble	✓	×	×	82.0	98.2	98.0	**89.1**
Ensemble	×	✓	×	72.0	97.6	97.3	87.2
Ensemble	×	×	✓	77.0	97.8	97.8	88.6
MobileNetV2	✓	×	×	77.0	97.6	98.0	**88.7**
MobileNetV2	×	✓	×	69.0	97.8	97.1	86.9
MobileNetV2	×	×	✓	78.0	98.1	97.6	88.6
ResNet50	✓	×	×	71.0	97.5	97.6	**87.9**
ResNet50	×	✓	×	69.0	97.5	97.7	86.8
ResNet50	×	×	✓	72.0	97.6	97.9	87.8
FcaNet	✓	×	×	68.0	97.5	97.4	**86.8**
FcaNet	×	✓	×	72.0	97.5	96.6	86.1
FcaNet	×	×	✓	75.0	97.5	97.1	86.3

Table 6 Ensemble method results on ISIC-2017.

(%)

Method	Keratosis classification				Melanoma classification				Average AUC
	Sen	Spe	Acc	AUC	Sen	Spe	Acc	AUC	
Ensemble-Stack-Part	68.4	90.2	86.9	92.2	53.9	91.7	84.3	89.4	91.2
Ensemble-Weight-Part	71.1	92.4	89.2	94.0	57.2	92.5	85.6	90.8	**92.4**
Ensemble-Vote-Part	68.7	91.1	87.7	92.9	46.5	92.3	83.4	89.6	91.5
Ensemble-Fuse-Part	68.2	90.6	86.6	91.9	46.3	92.0	83.2	89.1	90.5
Ensemble-Stack-Full	70.1	90.8	87.7	91.6	56.0	91.5	84.6	90.3	91.0
Ensemble-Weight-Full	68.9	91.4	88.0	92.8	43.6	93.2	83.5	91.0	**91.9**
Ensemble-Vote-Full	66.3	90.9	87.2	91.4	44.7	93.0	83.6	90.4	90.9
Ensemble-Fuse-Full	66.1	90.7	87.4	91.2	43.9	92.7	83.1	90.1	90.7
Ensemble-Stack-Cross	71.2	89.6	86.0	89.7	54.6	93.3	85.8	90.0	89.8
Ensemble-Weight-Cross	70.3	92.1	88.8	93.3	51.3	91.7	83.8	89.0	**91.5**
Ensemble-Vote-Cross	66.9	91.7	88.0	91.8	48.3	92.1	83.6	89.8	90.9
Ensemble-Fuse-Cross	70.0	91.3	86.1	91.6	49.5	92.4	84.2	88.7	90.2

Table 7 Ensemble method results on ISIC-2018.

(%)

Method	Sen. MEL	Spe	AUC	BAcc
Ensemble-Stack-Part	73.3	96.2	95.7	88.1
Ensemble-Weight-Part	82.8	98.2	98.0	**89.1**
Ensemble-Vote-Part	68.5	94.8	94.6	87.2
Ensemble-Fuse-Part	67.5	93.9	93.4	86.8
Ensemble-Stack-Full	69.3	96.3	94.7	86.5
Ensemble-Weight-Full	72.0	97.6	97.3	**87.2**
Ensemble-Vote-Full	62.4	97.8	93.8	83.6
Ensemble-Fuse-Full	63.9	97.3	94.3	84.2
Ensemble-Stack-Cross	70.8	95.2	95.3	87.3
Ensemble-Weight-Cross	74.0	95.8	96.1	**88.6**
Ensemble-Vote-Cross	77.0	97.8	97.8	86.9
Ensemble-Fuse-Cross	73.1	95.2	95.4	86.4

results demonstrate that the weight method outperforms the stack, vote and fuse methods. While in Table 6, the stack method slightly outperforms the vote method, the situation is reversed in Table 7. This difference may be attributed to the characteristics of the datasets. The feature fusion method shows no obvious advantages over other methods.

4.9 Visualization of CAM

CAM plays a vital role in highlighting the essential zones that demand more attention[5, 10]. It serves as an effective tool to enhance the model's performance. In this study, CAM assesses the impact of each branch on the weight block. Figure 2 illustrates CAM in three branches and the weights calculated by the weight block. Each branch possesses its image resolution, and

the selected zones are cropped out and resized to 224 pixel × 224 pixel. Consequently, smaller cropped zone sizes correspond to higher image resolutions. In Fig. 7, as we progress from the first row to the third row, image resolution increases. Generally, a model can extract layout information more effectively at lower resolutions. In some instances, images from all branches can be beneficial depending on the ratio of the disease zone in the image. Upon comparing disease images with CAM, it becomes evident that nearly all high-resolution branches can accurately pinpoint the key zones. W_1 represents the weight of the second row, and W_2 indicates the weight of the third row. The values of W_1 and W_2 signify the importance of each branch. Understanding the weight value mechanism is intricate. One plausible explanation is that when the disease zone covers the CAM region, it implies that the disease zone provides more supportive information, leading to a correct branch prediction and resulting in a higher weight value. Based on the findings presented in Tables 6 and 7, and Fig. 7, it can be concluded that the weight block mechanism indeed enhances the performance of the fused model.

4.10 Effect of λ

To achieve an optimal balance between two components in the part cross-entropy loss function, we introduce a parameter, denoted as λ, to fine-tune their relationship. We conduct classification experiments with varying values of λ, including 0.2, 0.4, 0.6, 0.8, 1.0, and 1.2, specifically for the Ensemble method. The results depicted in Fig. 8 illustrate that metrics such as AUC and BAcc exhibit changes in response to the value of λ, and there exists an optimal value for performance. Notably, the influence of λ on BAcc is more pronounced than on AUC. It is observed that when the value of λ is approximately 1, the model tends to achieve its best performance.

4.11 Complexity analysis

The number of FLOPs and parameters of our methods and other seven methods are listed in Table 8. We can find that the data of the fused models nearly equals the sum of the sub-models. The ensemble model has 8.11×10^9 FLOPs and 4.719×10^7 parameters. Compared to the MB-DCNN, GP-CNN-DTEL, and CI-Net, except for our MobileNetV2, the number of parameters of the fused model is close to or larger than them, but the value of FLOPs is lower than MB-DCNN, CP-CNN-DTEL, and CI-Net. It demonstrates that our model will cost more memory space, yet less computer source.

5 Ablation Study

We assess the effectiveness of our approach through an ablation study conducted on the ISIC2018 dataset, as presented in Table 9. The baseline model involves the fusion of ResNet50, MobileNetV2, and FacNet using stack branch fusion and cross loss. Upon careful analysis of the table, it is evident that all individual techniques, as well as their combinations, contribute to enhancing the model's performance. Specifically, Dynamic weight achieves a 1.3% increase in BAcc and a 3.2% improvement in Sen. MEL. Part loss results in a 0.8% enhancement in BAcc and a 2.5% improvement in Sen. MEL. The combined method leads to a 1.8% increase in BAcc and a 12% improvement in Sen. MEL.

6 Discussion

The experimental results highlight that MDP-DCNN has achieved notable performance compared to the state-of-the-art methods. The part cross-entropy loss function, by considering the influence of non-targeted

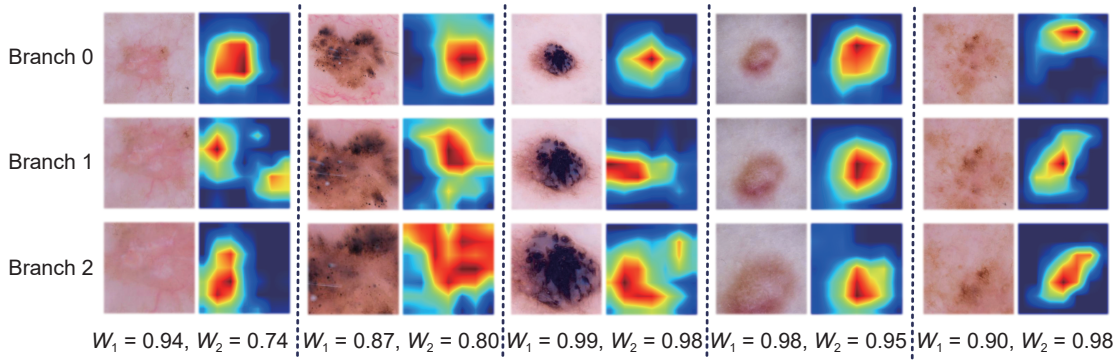


Fig. 7 Illustration of weight block. Each branch has a dynamic weight.

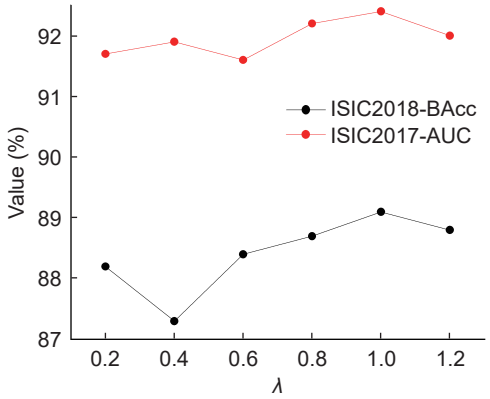


Fig. 8 Illustration of weight block. In the weight block, we reversely gray the cropped zone to highlight the disease area. Combining the CAM and the gray image as the inputs of two dilated convolutions.

Table 8 Complexity analysis.

Method	FLOPs ($\times 10^9$)	Number of params ($\times 10^6$)
MobileNetV2	0.32	2.23
ResNet50	4.12	23.52
FcaNet	3.67	21.45
CrossVIT[56]	3.50	54.54
MB-DCNN[8]	62.80	54.90
GP-CNN-DTEL[5]	31.80	37.20
CI-Net[55]	21.20	38.80
Ours (Ensemble)	8.11	47.19
Ours (MobileNetV2)	0.97	6.71
Ours (ResNet50)	12.41	70.56
Ours (FcaNet)	11.05	64.38

Table 9 Ablation study on ISIC-2018.

(%)

Dynamic weight	Part loss	Sen. MEL	Spe	AUC	BAcc
×	×	70.8	95.2	95.3	87.3
✓	×	74.0	95.8	96.1	88.6
×	✓	73.3	96.2	95.7	88.1
✓	✓	82.8	98.2	98.0	**89.1**

labels and focusing on wrong predictions, helps mitigate overfitting issues associated with cross-entropy loss. The current trend often involves using patch-based cropping methods with CAM to extract more information. While smaller patches aid in uncovering detailed information, they may also reduce discriminative features and harm prediction effectiveness. In addressing the limitations of patch-based methods, the proposed approach adopts a multi-scale branch with dynamic weight learning from CAM and grey images. The weight calculation in Fig. 4 involves complex interactions, with superposition between CAM and grey images potentially increasing the weight. However, the large size of the CAM may indicate lower reliability of the branch result. As of now, the proposed method uses fixed cropping sizes of 336 pixels and 224 pixels for the key zone. This approach only incorporates two scales, and these may not always be the optimal resolution candidates. Additionally, the number of branches is fixed and should be optimized.

7 Conclusion

Our proposed MDP-DCNN aims to overcome limitations observed in existing methods, specifically addressing fixed cropping zones and branch weights. MDP-DCNN comprises three main components: multiple-branch dynamic cropping, weight block, and part cross-entropy loss. The multiple-branch design allows for capturing more discriminative information, while dynamic cropping accurately locates key zones. The low-resolution branch focuses on global features like shape and symmetry, while high-resolution branches target detailed differences such as veins. The high-resolution image is derived from the low-resolution image cropped by Grad-CAM. Before fusing all branches, the weight block assesses the importance of high-resolution branches by analysing their Grad-CAM and gray images. The weight block employs dilated CNN to obtain a broader receptive field for evaluating CAM and gray images. To demonstrate the effectiveness of the proposed model, experiments are conducted on two public skin lesion datasets: ISIC-2017 and ISIC-2018. MDP-DCNN is compared with the best results on the ISIC live leaderboard and some state-of-the-art works. Results show that without the external dataset, MDP-DCNN achieves the second-place position on ISIC-2017 with an average AUC of 93.1% and on ISIC-2018 with BAcc (89.1%) and Sen. MEL (82.0%). We tested the influence of the sub-model type, fusion method, and loss function on results. MobileNetV2, ensemble models, and part loss perform better than others. Additionally, experiments to find the optimal value of the parameter λ indicate that the model performs better when λ equals 1.0. In terms of complexity analysis, our ensemble model has 8.11×10^9 FLOPs and 4.719×10^7 parameters. Compared to MB-DCNN, GP-CNN-DTEL, and CI-Net, except for our MobileNetV2, the number of parameters of the fused model is close to or larger than them, but the value of FLOPs is lower than MB-

DCNN, CP-CNN-DTEL, and CI-Net. The ablation study demonstrates that all single tricks and combined tricks can promote model performance. In the present work, the input size and the number of branches are fixed. In the future, we plan to explore the influence of cropping zone size and the optimal number of branches.

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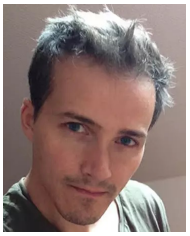
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